10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 5.2 — Unlike Denominators

Unit 5 | Solved Exercise | New Book (PECTAA)

Exercise 5.2 (Solved)

In this exercise the denominators are different. We take their LCM, write each fraction with the LCM as the common denominator, and then add or subtract the numerators. At the end, factorise and cancel if something is common.

1. Simplify.

(i) $\frac{3}{x-y} + \frac{1}{y-x}$

Since $y - x = -(x - y)$, the second fraction becomes $-1/(x - y)$

$= \frac{3}{x-y} - \frac{1}{x-y}$

Now the denominators are the same

$= \frac{2}{x-y}$

Answer: $2/(x - y)$

(ii) $\frac{4x}{x^2-3x+2} - \frac{4}{1-x} - \frac{5}{x-2}$

Factorise: $x^2 - 3x + 2 = (x - 1)(x - 2)$ and $1 - x = -(x - 1)$

$= \frac{4x}{(x-1)(x-2)} + \frac{4}{x-1} - \frac{5}{x-2}$

LCM of the denominators $= (x - 1)(x - 2)$. Taking LCM

$= \frac{4x + 4(x-2) - 5(x-1)}{(x-1)(x-2)}$

$= \frac{4x + 4x - 8 - 5x + 5}{(x-1)(x-2)}$

$= \frac{3x - 3}{(x-1)(x-2)}$

$= \frac{3(x-1)}{(x-1)(x-2)}$

Cancel $(x - 1)$

$= \frac{3}{x-2}$

Answer: $3/(x - 2)$

(iii) $\frac{x+y}{12x-6y} + \frac{x-y}{18x-9y}$

$12x - 6y = 6(2x - y)$ and $18x - 9y = 9(2x - y)$

$= \frac{x+y}{6(2x-y)} + \frac{x-y}{9(2x-y)}$

LCM of 6 and 9 is 18, so LCM of the denominators $= 18(2x - y)$

$= \frac{3(x+y) + 2(x-y)}{18(2x-y)}$

$= \frac{3x + 3y + 2x - 2y}{18(2x-y)}$

$= \frac{5x + y}{18(2x-y)}$

Answer: $(5x + y)/[18(2x - y)]$

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(iv) $\frac{2}{x+2y} - \frac{x-6y}{x^2-4y^2}$

$x^2 - 4y^2 = (x - 2y)(x + 2y)$, so the LCM is $(x - 2y)(x + 2y)$

$= \frac{2(x-2y) - (x-6y)}{(x-2y)(x+2y)}$

$= \frac{2x - 4y - x + 6y}{(x-2y)(x+2y)}$

$= \frac{x+2y}{(x-2y)(x+2y)}$

Cancel $(x + 2y)$

$= \frac{1}{x-2y}$

Answer: 1/(x − 2y)

(v) $\frac{1}{x+1} - \frac{2}{x+2} - \frac{2x+3}{x^2+3x+2}$

$x^2 + 3x + 2 = (x + 1)(x + 2)$, so the LCM is $(x + 1)(x + 2)$

$= \frac{(x+2) - 2(x+1) - (2x+3)}{(x+1)(x+2)}$

$= \frac{x + 2 - 2x - 2 - 2x - 3}{(x+1)(x+2)}$

$= \frac{-3x - 3}{(x+1)(x+2)}$

$= \frac{-3(x+1)}{(x+1)(x+2)}$

Cancel $(x + 1)$

$= -\frac{3}{x+2}$

Answer: −3/(x + 2)

(vi) $\frac{5}{x^2+x-6} - \frac{1}{2x^2-7x+6}$

$x^2 + x - 6 = (x + 3)(x - 2)$ and $2x^2 - 7x + 6 = (2x - 3)(x - 2)$

LCM $= (x + 3)(x - 2)(2x - 3)$. Taking LCM

$= \frac{5(2x-3) - (x+3)}{(x+3)(x-2)(2x-3)}$

$= \frac{10x - 15 - x - 3}{(x+3)(x-2)(2x-3)}$

$= \frac{9x - 18}{(x+3)(x-2)(2x-3)}$

$= \frac{9(x-2)}{(x+3)(x-2)(2x-3)}$

Cancel $(x - 2)$

$= \frac{9}{(x+3)(2x-3)}$

Answer: 9/[(x + 3)(2x − 3)]

(vii) $\frac{2x}{x+y} - \frac{y}{x-y} - \frac{2y^2}{x^2-y^2}$

$x^2 - y^2 = (x + y)(x - y)$, so the LCM is $(x + y)(x - y)$

$= \frac{2x(x-y) - y(x+y) - 2y^2}{(x+y)(x-y)}$

$= \frac{2x^2 - 2xy - xy - y^2 - 2y^2}{(x+y)(x-y)}$

$= \frac{2x^2 - 3xy - 3y^2}{(x+y)(x-y)}$

Answer: $(2x^2 - 3xy - 3y^2)$/[(x + y)(x − y)]

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(viii) $\frac{7}{2x^2 - x - 6} - \frac{8}{3x^2 - 4x - 4}$

$2x^2 - x - 6 = (2x + 3)(x - 2)$ and $3x^2 - 4x - 4 = (3x + 2)(x - 2)$

LCM $= (2x + 3)(3x + 2)(x - 2)$. Taking LCM

$= \frac{7(3x + 2) - 8(2x + 3)}{(2x + 3)(3x + 2)(x - 2)}$

$= \frac{21x + 14 - 16x - 24}{(2x + 3)(3x + 2)(x - 2)}$

$= \frac{5x - 10}{(2x + 3)(3x + 2)(x - 2)}$

$= \frac{5(x - 2)}{(2x + 3)(3x + 2)(x - 2)}$

Cancel $(x - 2)$

$= \frac{5}{(2x + 3)(3x + 2)}$

Answer: 5/[(2x + 3)(3x + 2)]

(ix) $\frac{x + 2}{x^2 - x - 12} - \frac{x}{x^2 + 6x + 9}$

$x^2 - x - 12 = (x - 4)(x + 3)$ and $x^2 + 6x + 9 = (x + 3)^2$

LCM $= (x - 4)(x + 3)^2$. Taking LCM

$= \frac{(x + 2)(x + 3) - x(x - 4)}{(x - 4)(x + 3)^2}$

$= \frac{x^2 + 5x + 6 - x^2 + 4x}{(x - 4)(x + 3)^2}$

$= \frac{9x + 6}{(x - 4)(x + 3)^2}$

$= \frac{3(3x + 2)}{(x - 4)(x + 3)^2}$

Answer: $3(3x + 2)/[(x - 4)(x + 3)^2]$

(x) $\frac{1}{x^2 - 4x + 3} + \frac{1}{x^2 - 3x + 2} - \frac{1}{x^2 - 5x + 6}$

The denominators are $(x - 1)(x - 3)$, $(x - 1)(x - 2)$ and $(x - 2)(x - 3)$

LCM $= (x - 1)(x - 2)(x - 3)$. Taking LCM

$= \frac{(x - 2) + (x - 3) - (x - 1)}{(x - 1)(x - 2)(x - 3)}$

$= \frac{x - 2 + x - 3 - x + 1}{(x - 1)(x - 2)(x - 3)}$

$= \frac{x - 4}{(x - 1)(x - 2)(x - 3)}$

Answer: $(x - 4)/[(x - 1)(x - 2)(x - 3)]$

(xi) $\frac{x^3}{x - y} + \frac{y^3}{y - x}$

Since $y - x = -(x - y)$

$= \frac{x^3}{x - y} - \frac{y^3}{x - y}$

$= \frac{x^3 - y^3}{x - y}$

$= \frac{(x - y)(x^2 + xy + y^2)}{x - y}$

Cancel $(x - y)$

$= x^2 + xy + y^2$

Answer: $x^2 + xy + y^2$

2. Subtract $1/(x^2 + 2)$ from $(2x^3 + x^2 + 3)/(x^2 + 2)^2$.

Write the two fractions one below the other, take $(x^2 + 2)^2$ as the common denominator, and subtract the numerators.

$$\frac{2x^3 + x^2 + 3}{(x^2 + 2)^2} - \frac{1}{x^2 + 2}$$
$$= \frac{(2x^3 + x^2 + 3) - (x^2 + 2)}{(x^2 + 2)^2}$$
$$= \frac{2x^3 + x^2 + 3 - x^2 - 2}{(x^2 + 2)^2}$$
$$= \frac{2x^3 + 1}{(x^2 + 2)^2}$$

Answer: $(2x^3 + 1)/(x^2 + 2)^2$

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